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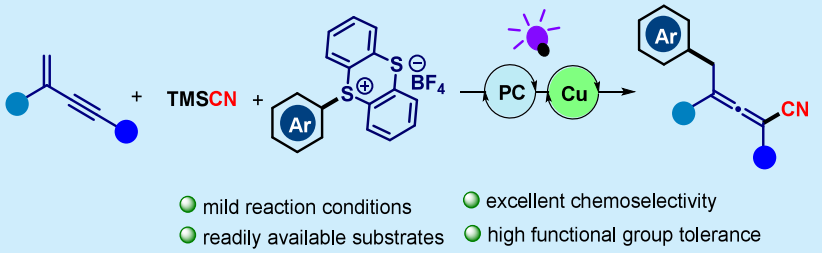
Photoredox/Copper Dual-Catalyzed Regioselective 1,4-Arylcyanation of 1,3-Enynes for Multifunctionalized Allenyl Nitriles

Qilong Wu, Xufeng Li, Jian Lv, Yongjia Shi, and Daoshan Yang*

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ABSTRACT: A copper/photoredox dual-catalysis-enabled 1,4-arylcyanation of 1,3-enynes to afford tetrasubstituted allenyl nitriles was initially developed. The reaction features good reaction generality and high reaction regioselectivity, providing facile access to multifunctionalized allenyl nitriles. This strategy achieves the construction of aryl-containing tetrasubstituted allenyl nitriles via indirect arene C−H functionalization. It also serves as an important complement to thianthrenium salt chemistry.

As an important class of molecules, allenes find widespread applications in natural products and pharmaceuticals (Scheme 1a).[1] Additionally, they play crucial roles in synthetic chemistry and drug design.[2] Therefore, chemists have been making persistent efforts to synthesize functionalized allenes including nucleophilic substitutions,[3] prototropic isomerization of alkynes,[4] sigmatropic rearrangement of propargylic molecules,[5] and other elegant methods.[6] Despite the success of these methods, the unavailability of prefunctionalized starting materials has limited their wide applications. In recent years, conjugated 1,3-enynes have increasingly been considered as key synthons for creating multisubstituted allenes, which is achieved through 1,4-addition to bring in two functional groups at the same time.[7] In this context, transition-metal-catalyzed 1,4-difunctionalization of 1,3-enynes through allenyl ionic intermediates is well developed (Scheme 1b).[8] Radical-induced 1,4-difunctionalization of 1,3-enynes can generate highly reactive and unusual short-lived allenyl radical intermediates, which can easily form allenyl cations or allenyl anions through oxidation or reduction pathways. In this regard, 1,4-difunctionalization of 1,3-enynes using a metallic radical catalysis strategy has emerged as a reliable robust approach toward functionalized allenes.[9] Very recently, a number of elegant synthetic methods have been developed by Ma,[10] Wu,[11] Liu,[12] Bao,[13] and other groups.[14] In these reactions, allenyl radicals were obtained via radical addition to the C=C double bond in 1,3-enynes and subsequent isomerization. The reactive allyl radicals reacted with the metal-FG (functional group) species, generating metal-allenyl intermediates that underwent reductive elimination to give the allenes (Scheme 1c). Although impressive success and breakthroughs have been achieved in this field, there are still challenges in controlling the regio- and chemoselectivities of multicomponent catalytic systems through the free radical approaches.

Organic reactions promoted by visible light are regarded as some of the best green methods.[15] Very recently, significant progress has been made in the synthesis of multifunctionalized allenyl nitriles through free radical 1,4-difunctionalization of 1,3-alkynals under photoredox/transition metal catalysis conditions. In these works, the radical precursors are mainly limited to C_{sp^3} sources (Scheme 1d).[16] Literature surveys indicated that 1,4-arylcyanation of 1,3-enynes to afford tetrasubstituted allenyl nitriles has not been reported thus far. The reason might be the lack of appropriate sources of aryl radicals and effective catalytic systems. Recently, thianthrenium salts have been widely employed as aryl radical precursors in diverse organic transformations.[17] Based on the excellent work reported previously and the continuous interest of our group in photocatalytic radical reactions,[18] we hypothesize that merging visible light photoredox catalysis and transition metal catalysis might offer a general strategy for synthesizing multifunctionalized allenyl nitriles through radical-type 1,4-difunctionalization of 1,3-enynes (Scheme 1e).

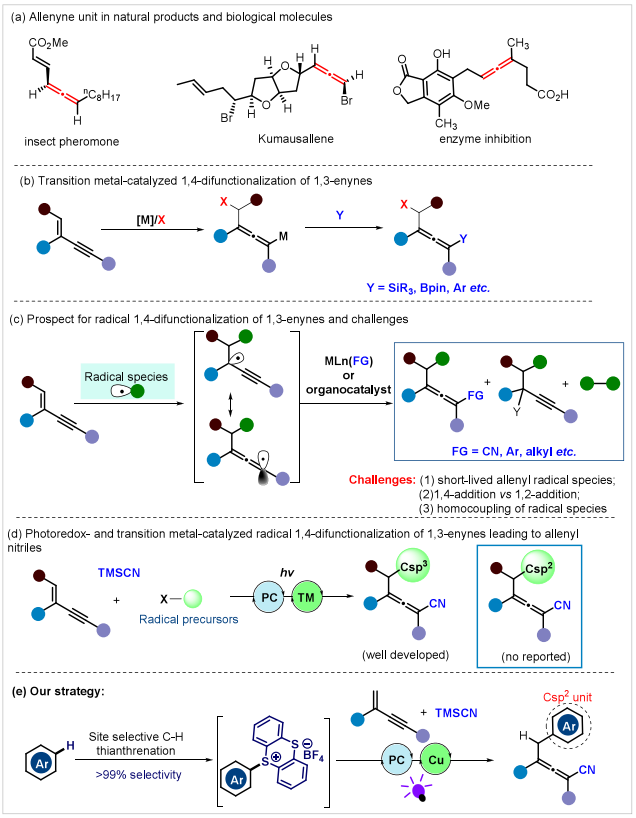
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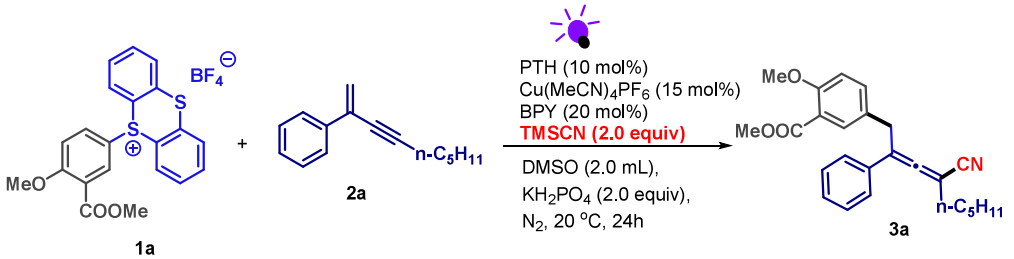
Scheme 1. Background and Our Strategy



At the outset of our investigation, 1,3-enyne **2a**, aryl thianthrenium salt **1a**, and TMSCN were chosen as the model substrates to optimize the reaction conditions, including the ligands, solvents, catalysts, bases, and photoredox catalysts. As shown in Table 1, after a series of systematic screenings of reaction parameters, a 1,4-arylcyanation reaction was achieved to deliver the expected multifunctionalized allenyl nitrile **3a** in 68% yield using photoredox/copper dual catalytic systems (entry 1). Results of our experiments revealed that other nitrogen ligands led to decreased yields of **3a** (entry 2, see the SI for more details). Replacing DMSO with other solvents, including MeCN and DMA, was less efficient (entry 3, see the SI for more details). Compared to other copper salts, such as $Cu(MeCN)_4BF_4$ and CuBr, $Cu(MeCN)_4PF_6$ was the best choice for this transformation (entry 4, see the SI for more details). Some commercially available photocatalysts such as *fac*-$Ir(ppy)_3$, 4-CzIPN, eosin Y, and PTH were tested under irradiation of various light source, and the combination of PTH and 390 nm wavelength exhibited the highest reaction activity (entry 5, see the SI for more details). Next, we employed other bases instead of KH_2PO_4, and the efficiency of the reaction did not improve (entry 6, see the SI for more details). Control experiments proved that light irradiation is indispensable for this reaction (entry 7).

After obtaining the optimum reaction conditions, we investigate the synthetic potential of this photoredox/copper-catalyzed strategy for the synthesis of tetrasubstituted allenyl nitriles (Scheme 2). First, various aryl thianthrenium salts were examined, and either electron-donating or electron-withdrawing groups on the aromatic ring were all compatible with this protocol, giving the desired products in moderate to good yield (**3a**–**3i**). The structures of the products can be confirmed through single-crystal X-ray diffraction (**3g**). Additionally, the heterocycle-containing aryl thianthrenium salts smoothly converted to the corresponding products in moderate yields (**3h** and **3i**). A series of functional groups, including methoxyl, ester, cyano, a C–Cl bond, and a C–Br bond, were tolerated, which could afford an opportunity for further modifications. Unfortunately, using alkenyl or alkyl thianthrenium salts could not provide the corresponding products (see SI for more details). Next, we turned our attention to test the functional groups tolerance contained in 1,3-enyne. For the C4-alkyl substituted substrates, we fixed the alkyl chain to C_5H_{11} or C_4H_9 while testing an array of C2-aryl substituents. The results show that when the naphthyl group was located at the C2 position, the yield slightly decreased (**3j**–**3l** and **3r**). In addition, diverse functional groups such as electron-withdrawing groups (Cl and I) and electron-donating

Table 1. Optimization of Reaction Conditions[a]

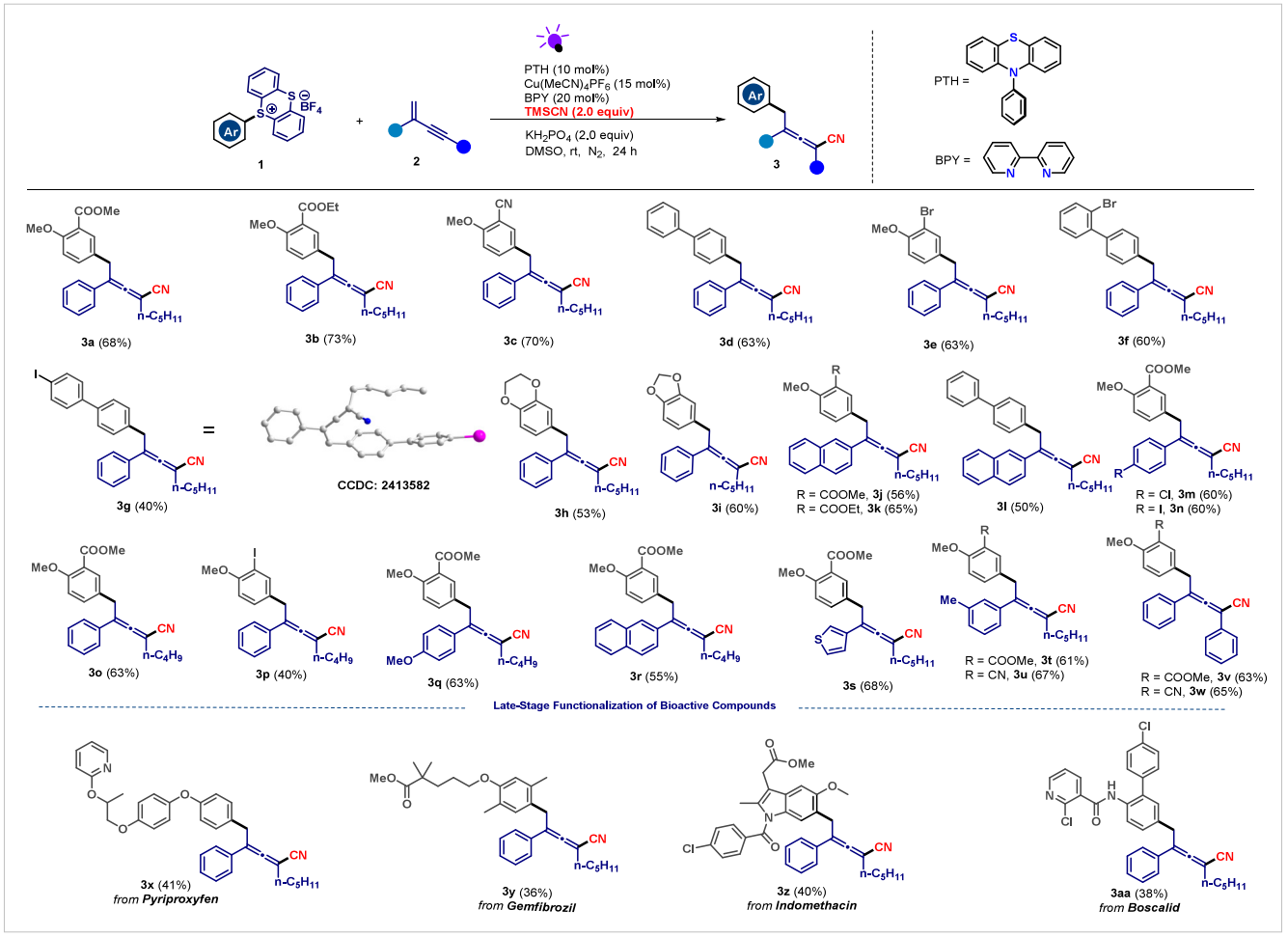


entry	variations from standard conditions	yield (%)[b]
1	none	68
2	ligands other than BPY	trace to 23
3	solvents other than DMSO	N.R. to 18
4	copper salts other than $Cu(MeCN)_4PF_6$	33–35
5	photoredox catalysts other than PTH	trace to 41
6	bases other than KH_2PO_4	34–37
7	no light	N.R.

[a]Reaction conditions: **1a** (0.3 mmol, 1.5 equiv), **2a** (0.2 mmol), TMSCN (0.4 mmol, 2.0 equiv), base (0.4 mmol, 2.0 equiv), photocatalyst (10 mol %), copper salt (15 mol %), ligand (20 mol %), and solvent (2.0 mL) at 20 °C under irradiation with a 15 W purple LED (390 nm) for 24 h. [b]Isolated yield. N.R. = no reaction.

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Scheme 2. Substrate Scope of Intermolecular 1,4-Arylcyanation of 1,3-Enynes[a]



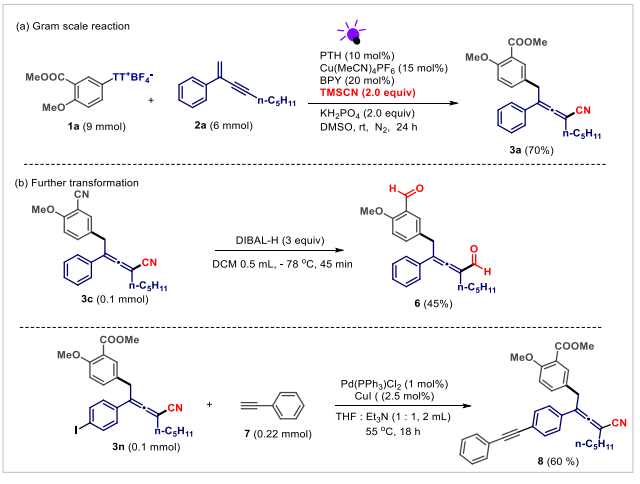
[a]Reaction conditions: **1** (0.3 mmol, 1.5 equiv), **2** (0.2 mmol), TMSCN (0.4 mmol, 2.0 equiv), KH_2PO_4 (0.4 mmol, 2.0 equiv), PTH (10 mol %), $Cu(MeCN)_4PF_6$ (15 mol %), BPY (20 mol %), and DMSO (2.0 mL) at 20 °C under irradiation with a 15 W purple LED (390 nm) for 24 h.

group (OMe) were all tolerated under the standard conditions. To our delight, when the C2 position was changed to thiophene, the desired product could also be obtained in good yield (**3s**). Unfortunately, when we changed the alkyl chain to C_6H_{13}, no desired product was observed (see SI for more details). Finally, numerous biologically active compounds, including pyriproxyfen, gemfibrozil, indomethacin, and boscalid, were well tolerated under the standard conditions and furnished the corresponding late stage-modified products (**3x**–**3aa**).

We carried out a gram-scale reaction to demonstrate the synthetic applicability of our method. As shown in Scheme 3a, the reaction proceeded smoothly and gave **3a** in 70% isolated yield under irradiation of two 390 nm LEDs (15 W), thereby indicating that this photochemical protocol is easy to scale up. To further demonstrate the practicability of the current method, the derivatization transformations of the obtained allenyl nitriles were conducted. As shown in Scheme 3b, the cyano group in product **3c** could be transferred to the aldehyde group at −78 °C with 3 equiv of DIBAL-H. Additionally, treatment of **3n** with phenylacetylene **7** provided the coupling product **8** in 60% yield through a palladium-catalyzed Sonogashira coupling reaction.

To gain an understanding of the mechanism, a series of control experiments were conducted. As shown in Scheme 4a,

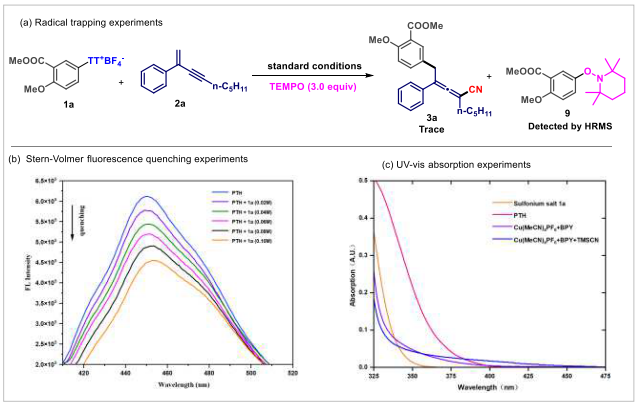
Scheme 3. Synthetic Applications



the model reaction was completely inhibited by adding 3.0 equiv of the radical scavenger (2,2,6,6-tetramethylpiperidin-1-yl)oxidanyl (TEMPO), and the TEMPO-captured **9** was detected by HRMS analysis. These results indicated that the aryl radical might be the key species in the current conversion process. Furthermore, Stern−Volmer fluorescence quenching

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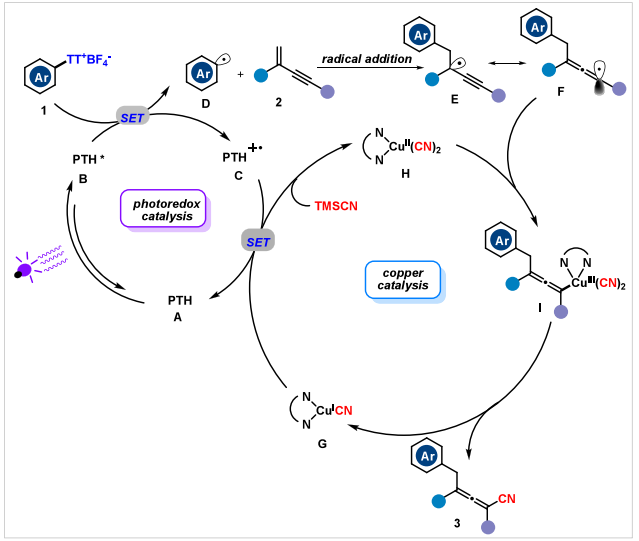
Scheme 4. Investigations of the Reaction Mechanism



experiments were conducted by mixing a solution of aryl thianthrenium salt **1a** and PTH (Scheme 4b). When the aryl thianthrenium salt **1a** was added in a quantitative manner, the fluorescence of the excited-state PTH gradually decreased, which proved that the PTH could be effectively quenched by aryl thianthrenium salts. Furthermore, UV−vis spectroscopy was also employed to determine the photoactive species involved in the current conversion process. The absorption of reaction components thianthrenium salt **1a**, PTH, Cu(MeCN)$_4$PF$_6$ + BPY, and Cu(MeCN)$_4$PF$_6$ + BPY + TMSCN were measured. As shown in Scheme 4c, only PTH showed an absorption band around the wavelength of the purple LED, thereby suggesting that PTH might serve as a photoredox catalyst during the process. Furthermore, cyclic voltammetry was used to determine the reduction potential of **1a** in DMSO (E_p = −1.51 V vs SCE), as well as the redox potential of of PTH ($E_{1/2}^{ox*}$ = −2.10 V vs SCE),[19] which indicated that the excited PTH could easily reduce the aryl thianthrenium salt **1a** (see the SI for details).

Based on the above experimental results, a hypothetical mechanism is depicted in Scheme 5. Excitation of photocatalyst PTH **A** leads to the active state **B**, which undergoes single electron transfer with the aryl thianthrenium salts **1** to produce the corresponding aryl radical **D**. Subsequently, the aryl radical adds to 1,3-enyne to form propargyl radical **E**, and its resonance structure is allenyl radical **F**. Subsequently, the allenyl radical **F** is trapped by LCu^{II}(CN)$_2$ **H** to produce the allenyl-Cu^{III} species **I**. Species **I** undergoes a reduction elimination reaction to generate a tetrasubstituted allenyl nitrile product **3**, while simultaneously regenerating Cu^{I} species **G**.

Scheme 5. Plausible Mechanism



In conclusion, we have successfully demonstrated the first photoredox/copper dual-catalysis-promoted 1,4-arylcyanation of 1,3-enynes for the synthesis of multifunctionalized allenyl nitriles using thianthrenium salts as aryl radical precursors. The reaction features exceptional regio- and chemoselectivities, mild reaction conditions, and broad functional-group tolerance and provides an efficient route to the synthesis of aryl-containing allenyl nitriles in a single step. Mechanistic studies indicate that aryl thianthrenium salts serve as aryl radical precursors and will inspire new opportunities for the design of radical-mediated multicomponent reactions. Further applications of this method and the catalytic asymmetric synthesis of optically active allenyl nitriles are currently underway.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c02449.

General experimental procedures, characterization data, and copies of ^{1}H NMR and ^{13}C NMR spectra for the products (PDF)

Accession Codes

Deposition Number 2413582 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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